
Tools & Technologies Used in V FOODS

A simple guide to the tools, libraries, and services that power V FOODS, explained in plain English.

React + Vite + TypeScript + Tailwind CSS

- **What it is:** The software used to build everything you see on the screen.
 - **What it does:**
 - **React:** Organizes the screens into reusable building blocks like menu cards, cart drawers, and live order trackers.
 - **Vite:** Bundles the code so pages load in less than a second, even on crowded college Wi-Fi or mobile data.
 - **TypeScript:** Catches spelling and data mistakes before the app is sent to students and shop staff.
 - **Tailwind CSS:** Styles every button, card, and layout cleanly so it looks great on any screen size.
 - **Why we chose it:** It creates a fast, app-like experience in any web browser without needing heavy downloads.
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Capacitor

- **What it is:** A tool that wraps the web application into an installable mobile app.
 - **What it does:**
 - Turns the website into an installable Android APK and iOS app.
 - Gives the app access to device features like camera scanning for QR codes at food counters.
 - **Why we chose it:** We only need to write the code once, and it works on web browsers, phones, and counter tablets alike.
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FastAPI (Python)

- **What it is:** A lightweight, fast backend engine running on our server.
 - **What it does:**
 - Handles secure communication between V FOODS, PhonePe, and SMS services.
 - Verifies digital signatures so nobody can fake a payment or top-up.
 - Triggers automated WhatsApp and SMS notifications when food is ready.
 - **Why we chose it:** It is lightweight, fast, and handles thousands of students ordering at the same break time without crashing.
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Supabase

- **What it is:** The secure cloud database and backend platform for V FOODS.
 - **What it does:**
 - Safely holds student wallets, shop menus, and live order queues.
 - Automatically updates kitchen screens instantly the moment an order is placed, with no page refreshing.
 - Runs strict security rules inside the database so users can only view their own data.
 - **Why we chose it:** It gives us bank-grade database security and instant live updates out of the box. Full details are in the [04-the-database.md](#) guide.
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PhonePe

- **What it is:** The official payment gateway used for all student wallet top-ups.
 - **What it does:**
 - Lets students load money into their campus wallet using any UPI app (such as PhonePe, Google Pay, or BHIM) or bank cards.
 - Sends a secure, digitally signed confirmation back to our server once money is received.
 - Zero fees on UPI/RuPay so students get 100% of their money credited to their balance.
 - **Why we chose it:** PhonePe is the most popular payment app among students and provides direct, reliable UPI settlement. It is used strictly for wallet top-ups; orders themselves are paid directly from the wallet balance.
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MSG91

- **What it is:** An enterprise communication service for sending SMS and WhatsApp alerts.
 - **What it does:**
 - Sends an instant SMS and WhatsApp message to the student when their food is marked "Ready for Pickup".
 - Sends one-time passwords (OTP) when users log in.
 - **Why we chose it:** It complies with Indian telecom regulations and delivers messages reliably in seconds, even if a student is sitting in a basement food court with weak mobile data.
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GitHub

- **What it is:** A cloud platform that safely stores the project's code.
- **What it does:**
 - Tracks every change made to the code over time, acting like an unlimited "undo" history.
 - Lets the team work on different features together without overwriting each other's work.
 - Connects directly to our hosting server so new code updates can be published automatically.
- **Why we chose it:** It is the industry standard for code storage, team collaboration, and version control.

Render

- **What it is:** The cloud hosting service that runs V FOODS on the live internet.
 - **What it does:**
 - Keeps the backend server and website running 24/7 at a public web address.
 - Automatically detects when new code is pushed to GitHub, builds the project, and deploys it live with zero downtime.
 - Provides automatic security certificates (HTTPS) so all student data is encrypted.
 - **Why we chose it:** It removes the headache of managing physical servers and keeps the app online automatically.
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Supabase MCP (Model Context Protocol)

- **What it is:** A secure bridge connecting the AI assistant to the live database during development.
 - **What it does:**
 - Lets the AI assistant check real table names, column structures, and stored procedures directly.
 - Verifies that code and database rules match reality without manual copying and pasting.
 - **Why we chose it:** It prevents guesswork and ensures all code changes match the real, running database.
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Figma MCP (Model Context Protocol)

- **What it is:** A bridge connecting the AI assistant to the original design files.
 - **What it does:**
 - Lets the AI inspect exact colors, spacing, and component designs straight from the design team.
 - Helps make sure the built screens match the approved mockups accurately.
 - **Why we chose it:** It ensures visual consistency and high design fidelity across all screens.
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